Car-Rental Management System (CRMS)

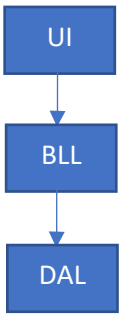
Overview

The **Car-Rental Management System** is a desktop application developed during my second year as a IT student. It is designed to streamline and digitize core car rental operations. The system provides intuitive workflows for managing vehicle fleets, customer profiles, rental transactions, maintenance logs, and basic user roles while adhering to clean software engineering principles and ensuring future scalability.

Technologies & Stack

- **Language:** C#
- **Framework:** .NET Framework (Windows Forms)
- **Data Access:** ADO.NET (Parameterized Queries / DataTables)
- **Database:** Microsoft SQL Server

Architecture

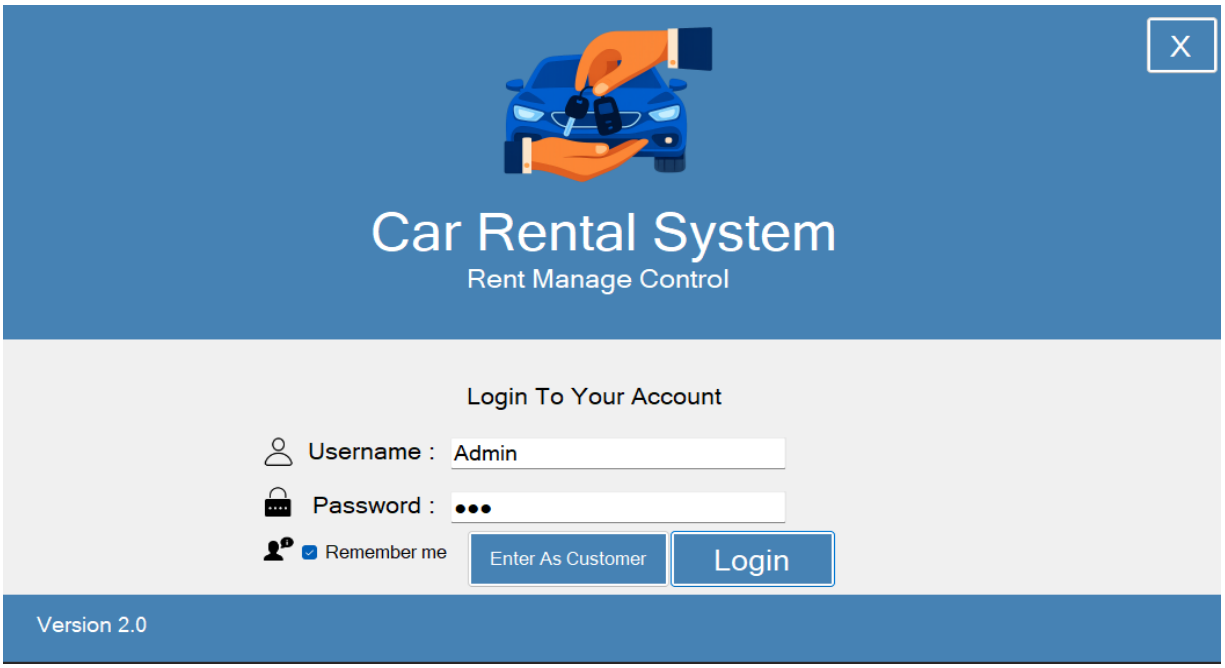


3-Tier Architecture: Structured with a strict separation of concerns across decoupled layers:

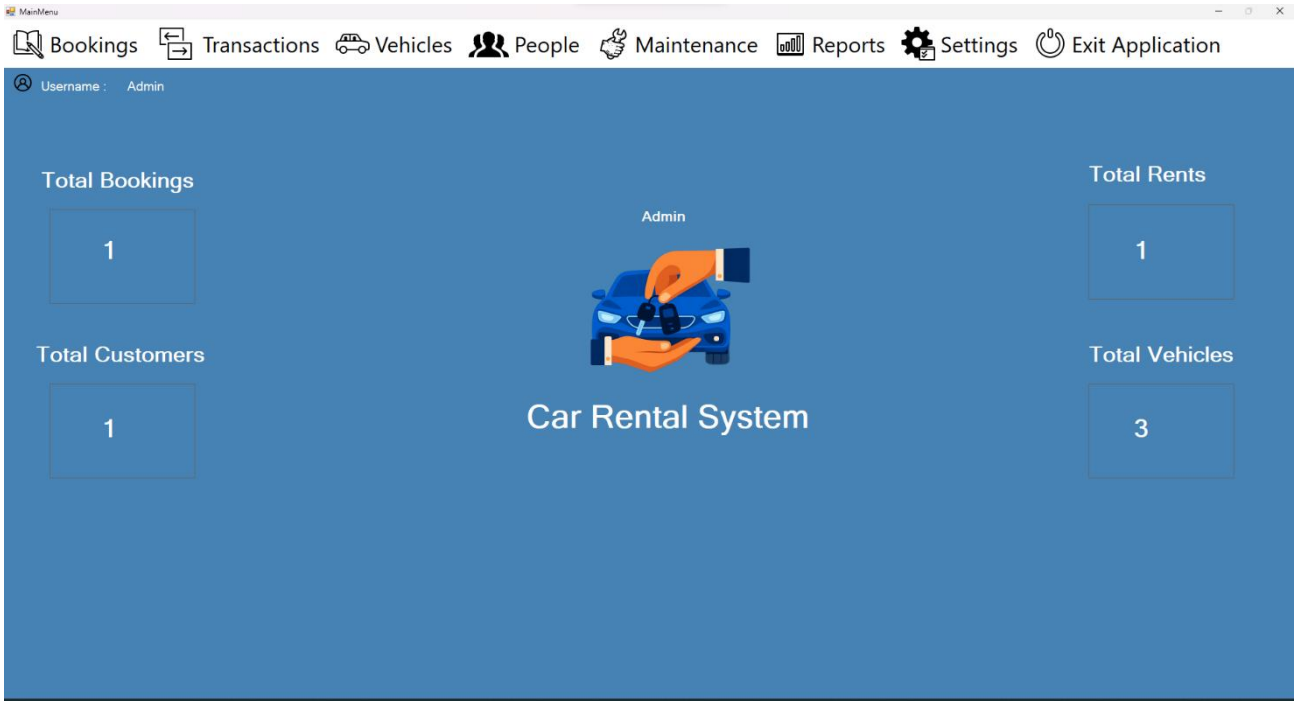
Main Features

1. **Fleet Management:** Manage vehicles categories,details and availability
2. **Rental Lifecycle Management:** Handle Bookings check-in/check-out and rent costs
3. **Vehicle Condition Tracking:** Basic recording of vehicle states before and after rentals, including mileage and fuel levels.
4. **Authentication & Security:** Secure user login utilizing password hashing .
5. **Bilingual Support:** Interface localization supporting seamless switching between Arabic and English.
6. **Basic Reporting:** Summary views displaying operational data such as fleet availability and popular vehicles.
7. **Customer Records:** Database for storing client contact details and tracking their individual rental histories.

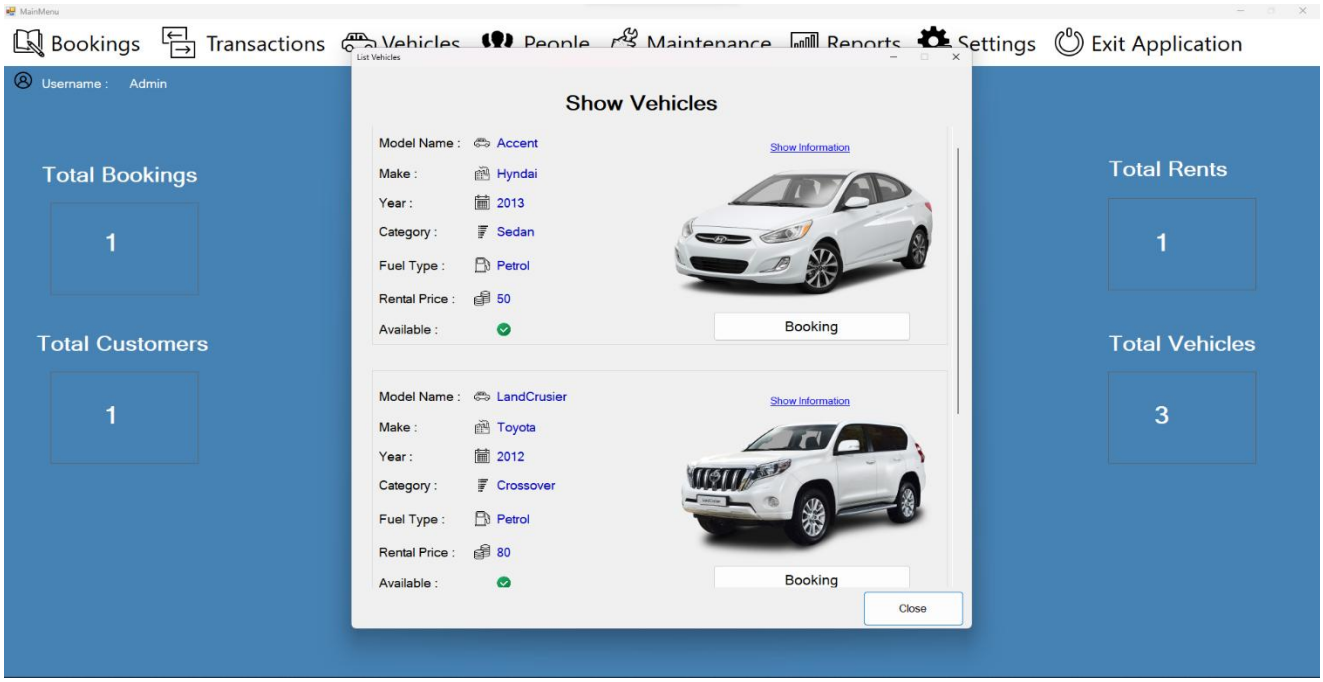
Photos & Screenshots



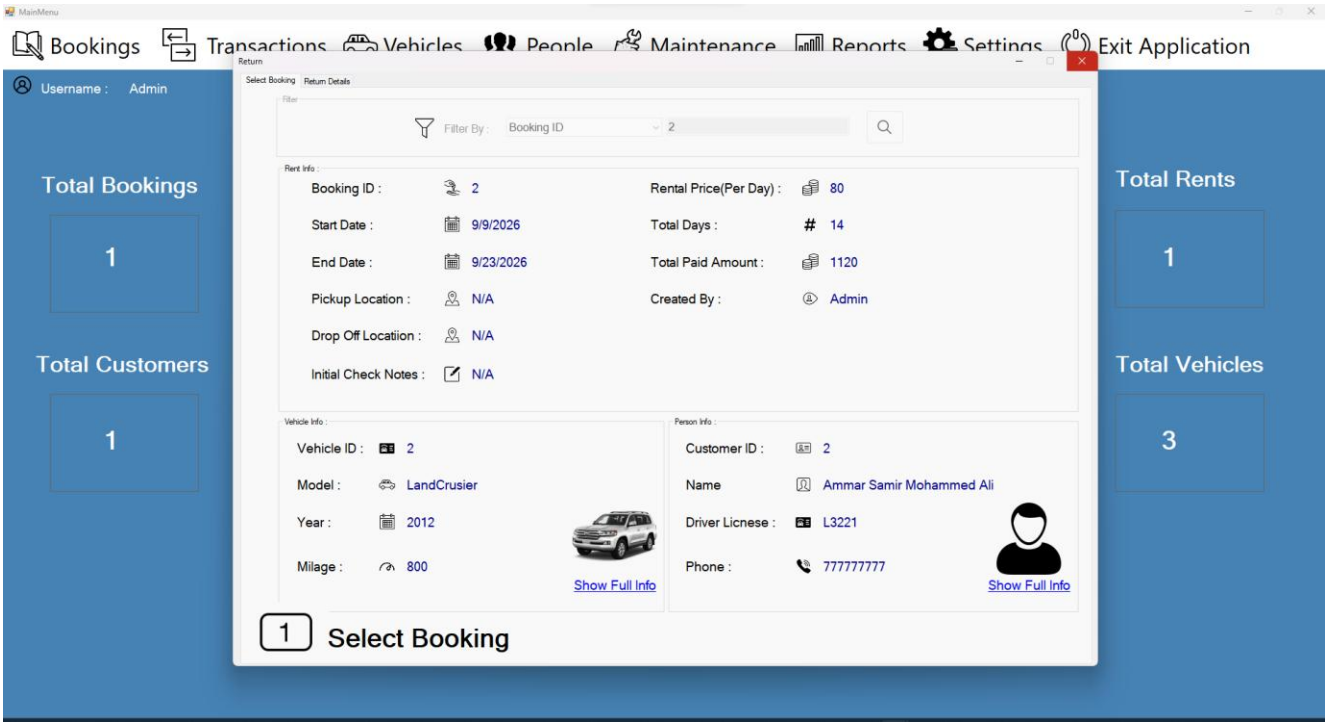
Login Form



Main Dashboard



Vehicle Show Form



Return Vehicle Form